PHANTOM

Coordinated listening evidence for investigation

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What it does

PHANTOM identifies statistically unusual coordinated listening structures within declared support. A population detector compares relational alignment with a conditional permutation reference; a separate graph core reconstructs members, targets, periods and supporting relationships. Evidence packages preserve alternative explanations and missing provenance for human review.

What the completed research shows

A source audit covered 704.7 million Deezer events; detector experiments used bounded populations. Controlled external validation transferred to ListenBrainz. In a later prospective final test, frozen before final population access, 4,032 listeners supplied 3,714,102 observed September events. The frozen detector achieved block and pooled AUC 1.000: all 378 planted treatment worlds alerted and none of 378 matched controls did.

These are controlled intervention labels on real listening histories. They do not measure real fraud accuracy, and zero observed control alerts does not establish zero natural-population alert probability. Separate historical internal graph testing achieved member and target F1 about 0.9167; the final population test does not externally validate the entire investigation stack.

Why the failures matter

Early benchmarks contained origin, session, catalog and duplicate shortcuts. Raw thresholds and attempted behavioral fusion failed. Shared-stimulus shocks and shifted overlap defeated broader robustness requirements. Different hidden causes can produce identical event histories, so no event-only score can universally establish manipulative intent. PHANTOM presents evidence and ambiguity rather than enforcement judgments.

Verification and provenance

One canonical prospective evaluation was followed by deterministic replay covering all 756 worlds. The original verifier failed while serializing its receipt because of a non-scientific type-conversion defect. That failure remains preserved. A separate post-final verifier confirmed every provisional metric from immutable original/replay artifacts without rerunning or modifying the scientific evaluation. Pre-access protocol and execution-lock corrections are documented in the paper.

What an investigator would need next

Check support, inspect temporal and graph evidence, compare alternatives and seek independently authorized exposure, referral or coordination provenance. The read-only dashboard demonstrates five synthetic runs and 22 hypotheses; analyst benefit has not been measured. No production deployment, real fraud labels or enforcement validation is claimed.

Release and responsibility

The technical report, supplement, aggregate figures and synthetic demo are prepared for author review. Raw datasets, protected truth and private logs are excluded. AI-assisted tools supported selected engineering tasks, with minor assistance in documentation and manuscript preparation. Nischit Girish Kumar directed the research, system architecture, experimental design, validation, and final technical decisions and assumes responsibility for the work. Independent research; not peer reviewed.